

Operating map (amendment A1: fixed threshold)

Mode 1

catalyst 1%	X	X	X	X	X	X	X	X
catalyst 2%	X	X	X	X	X	X	X	X
catalyst 5%	X	D	X	D	X	D	D	D
catalyst 10%	X	D	X	D	X	D	D	D
catalyst 20%								
membrane 1%	X	X	X	X	X	X	X	X
membrane 2%	X	D	X	D	X	D	X	D
membrane 5%	X	D	X	D	X	D	X	D
membrane 10%	X	D	X	D	X	D	X	D
membrane 20%	X	D	X	D	X	D	X	D
feed shortfall 1%	X	X	X	X	X	X	X	X
feed shortfall 2%	X	X	X	X	X	X	X	X
feed shortfall 5%	X	D	X	D	X	D	D	D
feed shortfall 10%	X	D	X	D	D	D	D	D
feed shortfall 20%	X	D	X	D	D	D	D	D
H2 sensor drift 1%	X	X	X	X	X	X	X	X
H2 sensor drift 2%	X	X	X	X	X	X	X	X
H2 sensor drift 5%	X	X	X	X	X	D	X	D
H2 sensor drift 10%	X	D	X	D	X	D	X	D
H2 sensor drift 20%	X	D	X	D	X	D	X	D
	S1	S1+E	S2	S2+E	S3	S3+E	S4	S4+E

Mode 2

	X	X	X	X	X	X	X	X
	X	s	X	s	X	s	X	s
	X	s	X	s	X	s	X	s
	X	s	X	s	X	s	X	s
	X	s	X	s	X	s	X	s
	X	s	X	s	X	s	X	D
	X	D	X	D	X	D	X	D
	X	D	X	D	X	D	X	D
	X	X	X	X	X	X	X	X
	X	D	X	D	X	D	X	D
	X	D	X	D	X	D	X	D
	X	D	X	D	X	D	X	D
	S1	S1+E	S2	S2+E	S3	S3+E	S4	S4+E

true cause and H2 drop at constant feed

D = distinguishable s = indistinguishable, same action X = indistinguishable, different action blank = not reachable
S1 = H2 flow; S2 = +outlet T; S3 = +waste flow; S4 = +micro-GC mole fractions; +E = bounded feed move (+0.0003 mol/min, 5 min)